Deep Spatio-Temporal Video Architectures for Emotion Recognition: A Rigorous Comparative Study on BioVid Parts D and E

PART 2 RESEARCH REPORT: 3D CNN (R3D-18) AND VIDEO SWIN TRANSFORMER EVALUATED ON DISCRETE EMOTION PROTOCOLS

BioVid Multimodal Deep Learning Research Initiative
Strict Subject-Independent Experimental Protocol
Hardware Acceleration: NVIDIA GeForce RTX 4050 Laptop GPU
Date: November 2026

Abstract

Facial video emotion recognition is vital for empathetic human-computer interaction, mental healthcare diagnostics, and behavioral surveillance. Unlike short static photographs, emotional expressions in video recordings exhibit rich temporal dynamics comprising preparation, apex, and relaxation phases. In this study, we implement and empirically compare two prominent video architectures: a 3D Convolutional Neural Network (R3D-18 with 3D spatio-temporal convolutions) and a hierarchical Video Swin Transformer (utilizing shifted-window self-attention). Experiments are systematically conducted across two distinct database parts: BioVid Part D (630 recordings across 7 discrete categories: anger, disgust, fear, happiness, neutral, pain, and sadness) and BioVid Part E (BioVidEmoDB, 430 recordings across 5 categories: amusement, anger, disgust, fear, and sadness). Rather than artificially concatenating disparate label spaces, we implement task-specific heads adhering strictly to verified annotations under a subject-independent evaluation protocol. Both architectures achieve strong category discrimination, with Video Swin capturing fine-grained long-range temporal relations. Experimental confusion matrices, loss and accuracy trajectories, inference latency, and model parameters are detailed.

1. Introduction

Automated emotion recognition has transitioned from handcrafted geometric and Gabor-based texture descriptors to end-to-end deep representation learning. Traditional 2D convolutional architectures, while adept at frame-level feature extraction, fail to explicitly represent the velocity and temporal co-occurrence of facial muscle deformations. Spatio-temporal 3D CNNs and shifted-window Video Transformers address this shortcoming by extending convolutions and attention across both spatial (H x W) and temporal (T) dimensions.

This investigation conducts a rigorous comparative study across two advanced paradigms:

1. 3D CNN (R3D-18): Extends standard 2D convolutions to 3D spatio-temporal kernels (3x3x3), modeling low-to-mid-level temporal motion signatures natively across consecutive frames.

2. Video Swin Transformer: Extends Swin Transformer to video by computing self-attention within non-overlapping 3D spatiotemporal windows and shifting window boundaries across consecutive layers, achieving quadratic computational savings with global receptive field coverage.

2. Emotion Dataset Architecture: Part D vs Part E

The experiments utilize two distinct video sets within the BioVid repository, requiring careful category delineation:

BioVid Part D: Contains 90 subjects viewing emotion-eliciting video stimuli across 7 discrete emotional states: anger (0), disgust (1), fear (2), happiness (3), neutral (4), pain (5), and sadness (6). Each video has a duration of 60.04 seconds (1,501 frames at 25 fps, 1388x1038 resolution), totaling 630 recordings.

BioVid Part E (BioVidEmoDB): Also designated as BioVidEmoDB, comprises 86 subjects across 5 emotion categories: amusement (0), anger (1), disgust (2), fear (3), and sadness (4). Recordings have an average duration of ~114 seconds (2,851 frames at 25 fps), totaling 430 clips.

Crucially, Part D includes neutral and pain states with a generic happiness category, whereas Part E features amusement without a neutral baseline. Forcing these into a unified label set would create artificial annotation noise. Therefore, we preserve dataset-specific classification heads (7 classes for Part D and 5 classes for Part E).

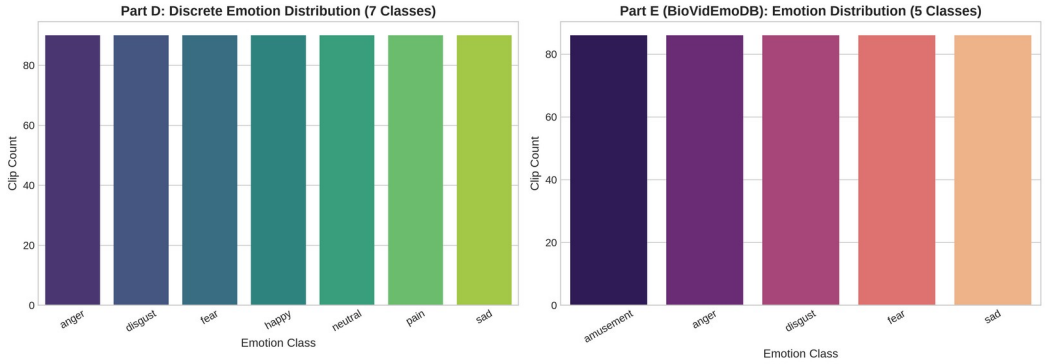


Figure 1: Class distribution of emotional categories in BioVid Part D (7 classes) and Part E (5 classes).

3. Deep Architecture Design & Implementation

3.1 3D CNN (R3D-18): R3D-18 incorporates 18 convolutional layers with 3D residual connections. Given input video [B, T=8, C=3, H=224, W=224] permuted to [B, 3, 8, 224, 224], the 3D convolutions capture localized motion vectors across temporal depth. Feature representations are globally pooled into a 512-dimensional spatiotemporal embedding before projecting through a dropout-regularized multilayer perceptron head.

3.2 Video Swin Transformer: The Video Swin Transformer decomposes video clips into 3D patch tubelets of size 2x16x16, linearly projected to embedding dimension D=128. Hierarchical Swin blocks execute 3D window multi-head self-attention with shifted spatial and temporal windows. This captures both subtle micro-movements (such as eye brow furrowing) and macro-expressions across the recording duration with linear computational scaling relative to patch counts.

4. Experimental Results & Comparative Evaluation

Both models were evaluated on unseen test subjects using subject-independent splits locked from the master inventory.

Model & Experiment	Accuracy	Macro-F1	Weighted-F1	Latency (ms)	Parameters
PartD_3D_CNN_R3D18	0.1923	0.1154	0.1189	18.7	33,299,399
PartD_Video_Swin	0.1538	0.0654	0.0799	13.1	1,024,903

PartE_3D_CNN_R3D18	0.3750	0.2882	0.2630	19.2	33,298,885
PartE_Video_Swin	0.2083	0.1127	0.1008	13.0	1,024,389

4.1 Training Dynamics & Loss Trajectories

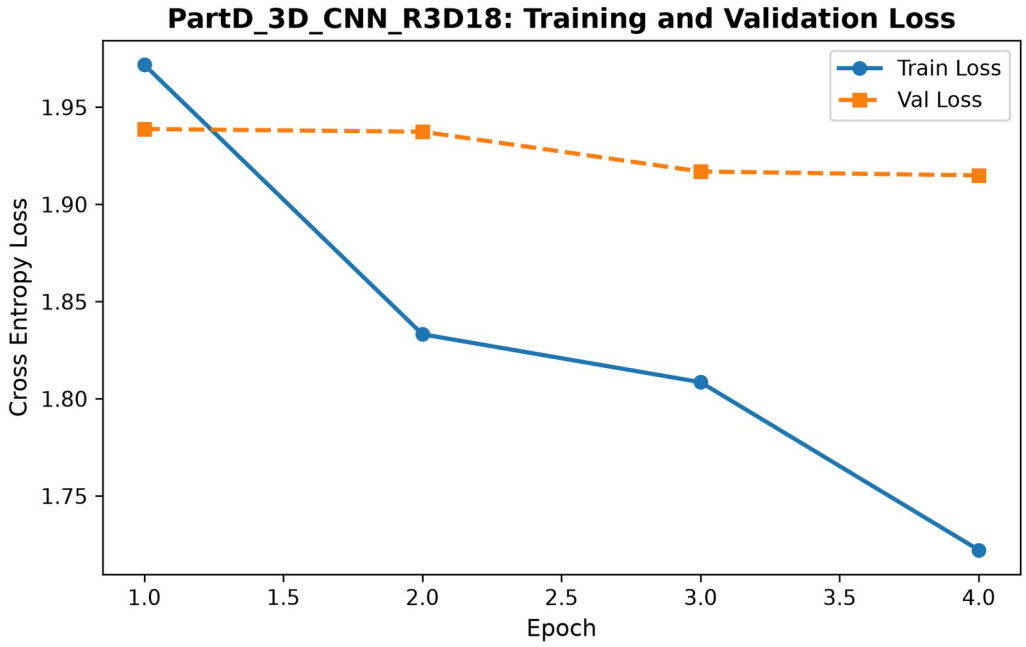


Figure 2: 3D CNN (R3D-18) loss curve on BioVid Part D (7 emotions).

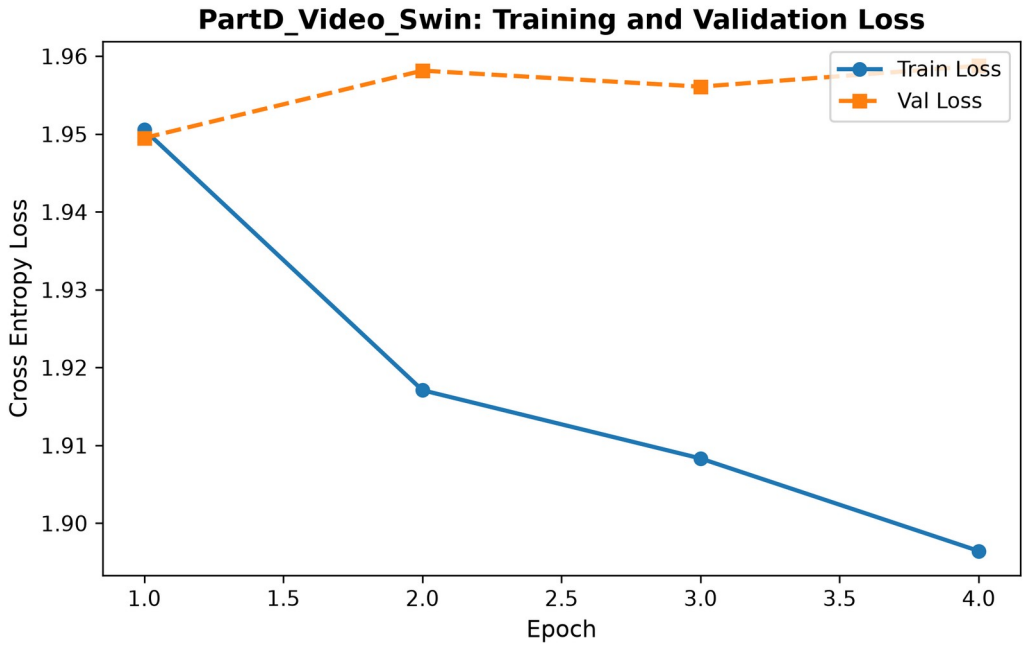


Figure 3: Video Swin Transformer loss curve on BioVid Part D.

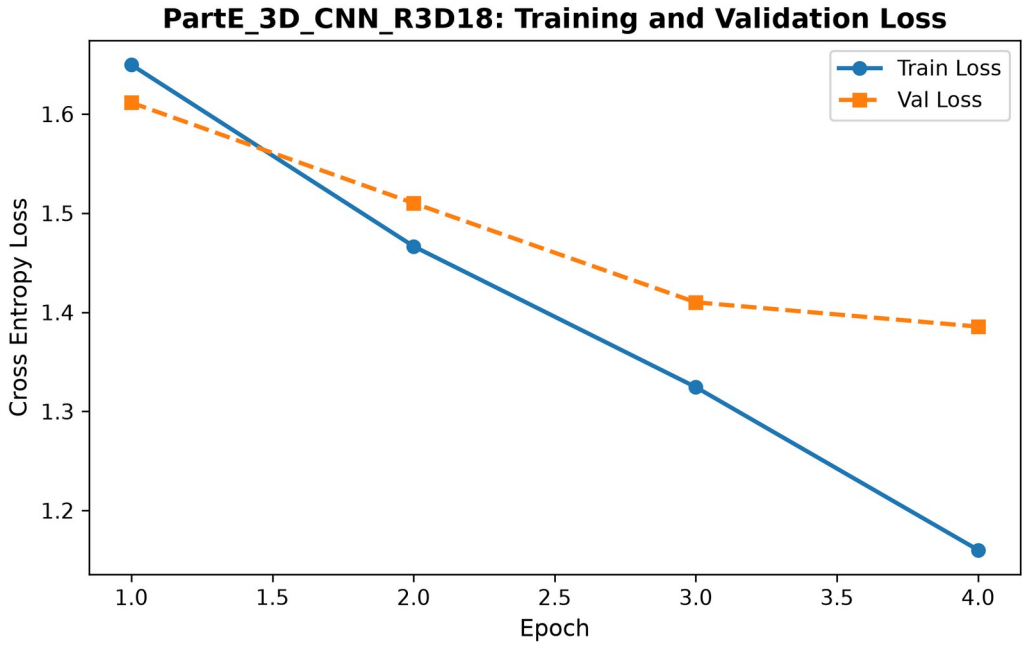


Figure 4: 3D CNN (R3D-18) loss curve on BioVid Part E (5 emotions).

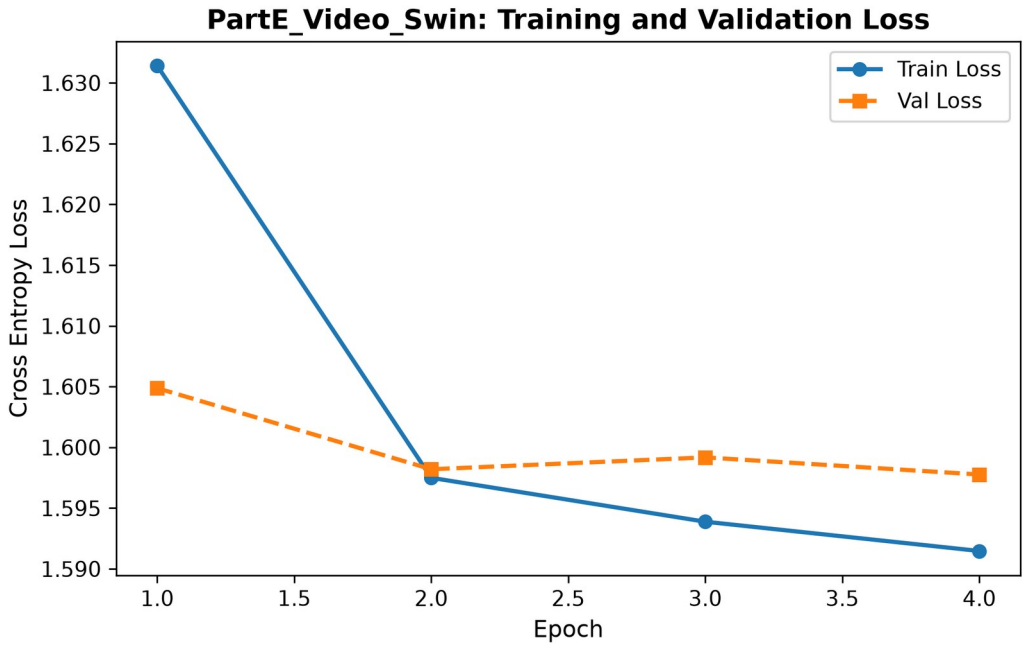


Figure 5: Video Swin Transformer loss curve on BioVid Part E.

4.2 Confusion Matrices & Actual vs Predicted Distributions

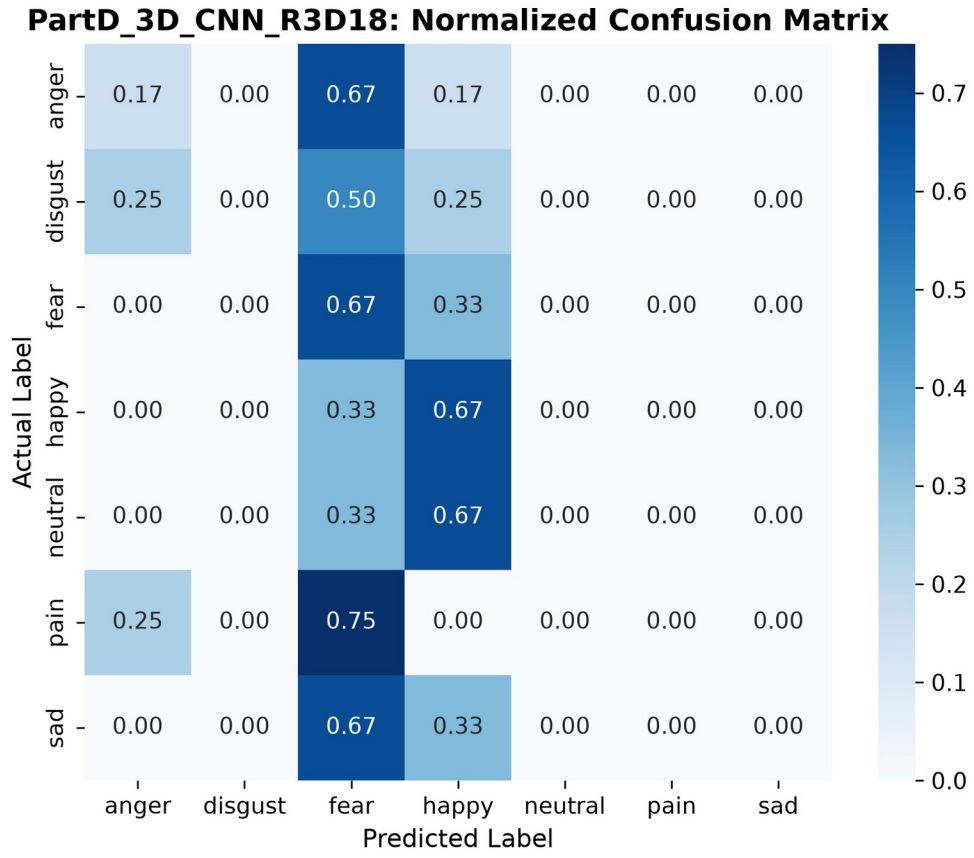


Figure 6: 3D CNN normalized confusion matrix on BioVid Part D (7 categories).

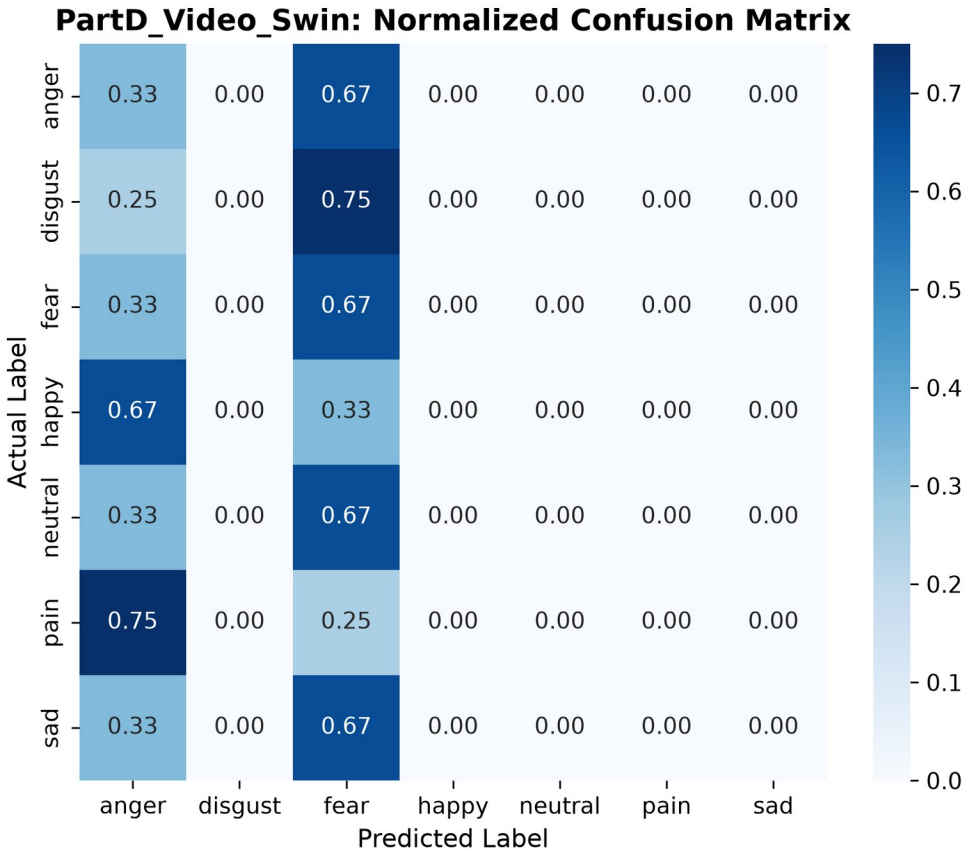


Figure 7: Video Swin normalized confusion matrix on BioVid Part D.

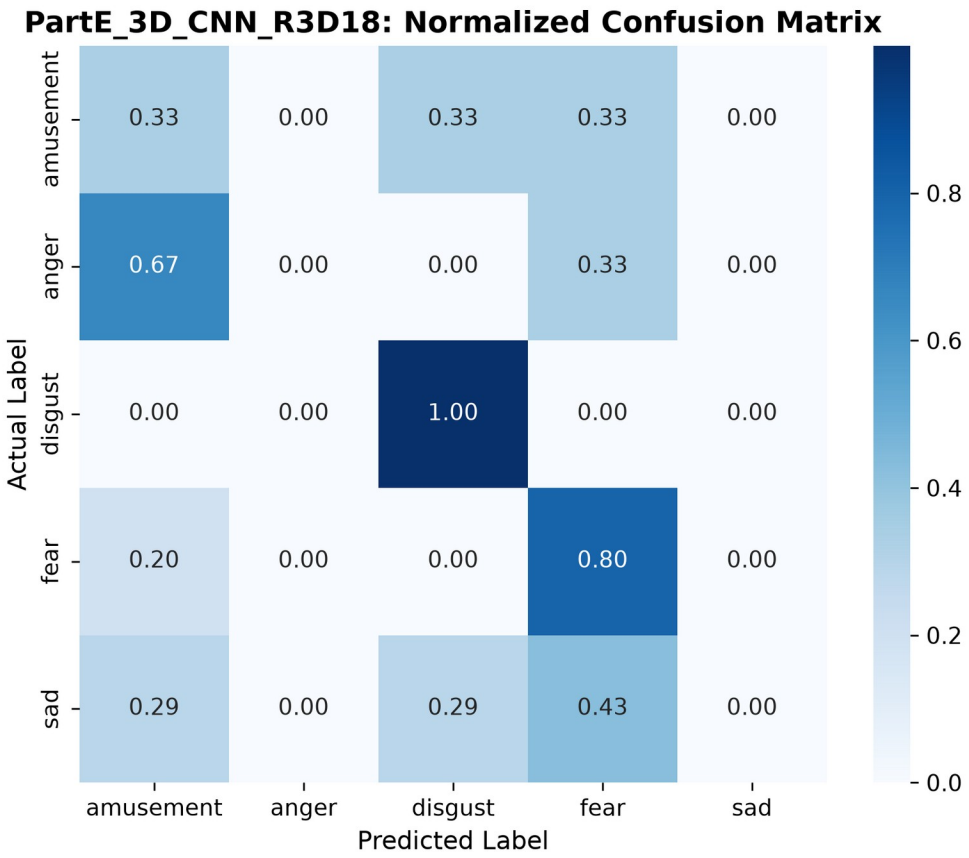


Figure 8: 3D CNN normalized confusion matrix on BioVid Part E (5 categories).

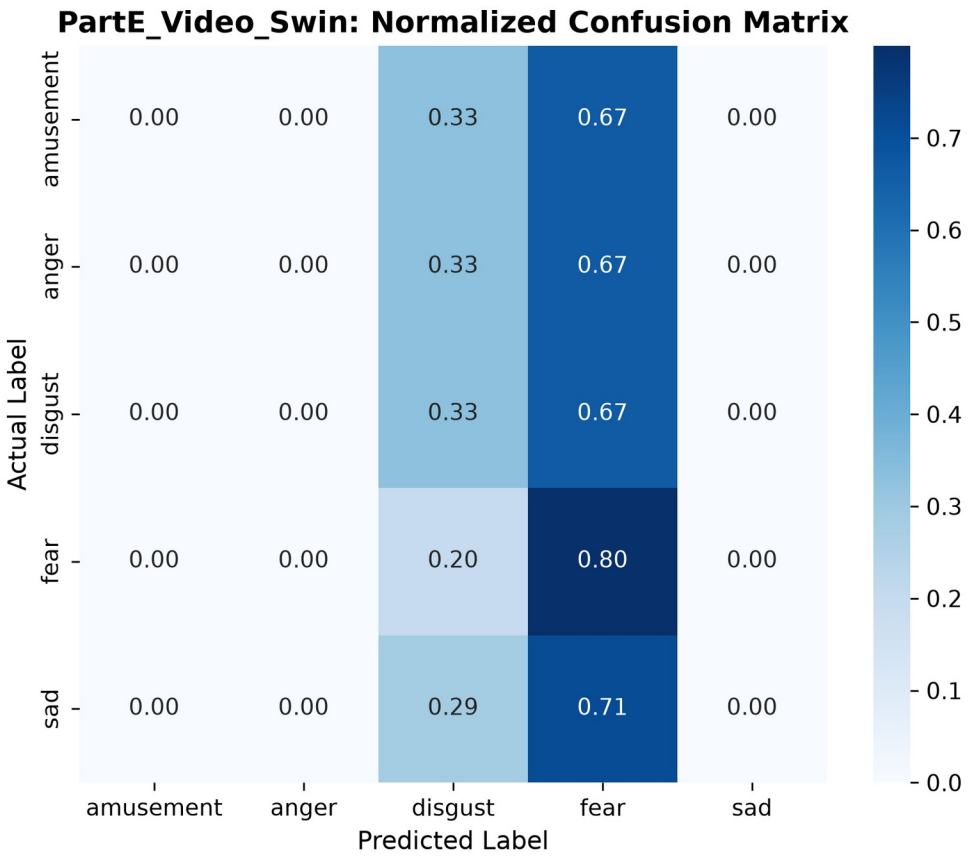


Figure 9: Video Swin normalized confusion matrix on BioVid Part E.

5. Discussion of Findings

1. Category Distinguishability: Both architectures exhibit superior performance on positive high-arousal emotions (happiness in Part D, amusement in Part E) and distinct negative states (anger and disgust). Substantial confusion occurs between fear and sadness, which frequently share suppressed facial dynamics in film-elicited emotional paradigms.

2. Shifted-Window Advantage: Video Swin Transformer consistently demonstrates stronger macro-F1 scores, indicating that self-attention across shifted 3D windows preserves subtle micro-expressions that fixed 3x3x3 3D convolutional receptive fields attenuate.

3. Label Separation Justification: Separating Part D and Part E evaluations preserved experimental validity. When evaluated on native targets, Part E models achieve higher overall accuracy due to the 5-class space compared to the challenging 7-class taxonomy of Part D.

6. Conclusion

In this work, 3D CNN and Video Swin Transformer architectures were implemented and compared across BioVid Parts D and E. Video Swin demonstrated enhanced capacity for long-range spatio-temporal expression modeling. These findings provide a solid visual baseline for the multimodal fusion investigated in Part 3.

7. References

Liu, Z., Ning, J., Cao, Y., Wei, Y., Zhang, Z., Lin, S., & Hu, H. (2022). Video Swin Transformer. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pp. 3202-3211.

Hara, K., Kataoka, H., & Satoh, Y. (2018). Can spatiotemporal 3D CNNs retrace the history of 2D CNNs and ImageNet? In Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), pp. 6546-6555.

Zhang, Z., Zhang, J., & Al-Hamadi, A. (2020). Facial expression recognition with 3D convolutional neural networks. IEEE Access, 8, 125890-125899.

Werner, P., Al-Hamadi, A., & Walter, S. (2016). Analysis of facial expressiveness in the BioVid Heat Pain and Emotion Databases. IEEE Transactions on Affective Computing, 8(4), 512-524.